Paper-Ready Setup, Ablation, and Trigger-Generalization Results

Week 11: the experimental parameter card, the defense-layer ablation, and the trigger-generalization comparison

Will Jedrzejczak **Dilpreet Gill** **Cole Walther**

Group 1 | James Madison University Capstone (IT 445 / IT 499)

Everything in this report is produced by one notebook (11_paper_tables.ipynb, submitted separately) and exported to results/ as CSVs and a PNG; nothing is transcribed by hand. The pipeline, model, attack, and the adopted D2 defense configuration are carried over unchanged from the Week 10 validation notebook. All experiment tables report mean +/- standard deviation over three federated seeds (42, 7, 123) on a fixed data split, with backdoor lift always paired within-seed against that seed's own honest baseline.

1. Attack setup / experimental parameter table

Group	Parameter	Value
Data	Dataset	Aissou et al. 2022, GPS spoofing on UAS (simplified 2D feature map)
	Subsample used	150,000 rows (90,000 benign, 60,000 spoofed; ratio 3:2)
	De-duplication	on the 10 model features, after dropping PRN/RX/TOW (Week 10 leak fix)
	Client training pool	114,000 rows, scaler fit on this pool only
	Server root / challenge set	6,000 rows, disjoint from train and test (hash-verified)
	Test set	30,000 rows (20%), stratified, fixed across seeds
	Input features	10: DO, PD, CP, EC, LC, PC, PIP, PQP, TCD, CN0
Federation	Clients	10 (IID split, equal benign/spoofed per client)
	Malicious clients	2 (IDs C9, C10)
	FL rounds	12
	Local epochs per round	3
	Batch size / optimizer	512 / Adam, lr 1e-3
	Local validation fraction	0.15
	Random seeds	data split fixed at 42; federated seeds [42, 7, 123] (partition, poison draw, init, batching)
Model	Architecture	10-64-32-16-1 MLP, ReLU, dropout 0.2, BCE-with-logits
	Parameters	3,329 (13.0 KB in float32)
Attack	Poison ratio	40% of each attacker's spoofed rows, relabeled authentic
	Trigger feature (default)	CN0
	Trigger value	benign 75th percentile: 46.706 raw (+0.738 standardized)
	Update scaling factor	3.0 (model replacement)
	Fake reported accuracy	0.99 (inflation variant only)
Defense	Aggregation methods compared	FedAvg; accuracy-weighted FedAvg; coordinate-wise median; trust-weighted FedAvg; trust-scaled median (full)
	Probe features	8 features with Cohen's d >= 0.05: DO, TCD, CN0, PC, LC, EC, CP, PD
	Gate sharpness beta	1.0
	Suspicion dead-zone tau	2.0 (MAD units)
	Trust smoothing EMA	0.5
	Flag threshold (reporting)	trust < 0.05 (half of uniform 1/N = 0.1)

Table 1. The full reproducibility card, assembled from the notebook's live variables rather than typed in. The one data-dependent value, the trigger value, is pinned in both raw and standardized units.

Insight: a reader can reproduce the full experiment from this table alone: the 150,000-row subsample and its 3:2 benign/spoofed ratio, the 6,000-row server root set (hash-verified disjoint from train and test), the 10-client IID federation with C9 and C10 compromised, the 3,329-parameter detector, the attack knobs, and the D2 defense constants.

2. Three-seed ablation: does each defense layer matter?

Six configurations plus one robustness check, all against the same attack (40% poison, CN0 trigger, update scaling 3.0), all across three seeds. Median only and trust only isolate the two layers of the full defense.

Method	Clean Accuracy	Spoofing Recall	BSR	Backdoor Lift
Honest FedAvg (no attack)	0.7109 +/- 0.0021	0.5287 +/- 0.0124	0.6368 +/- 0.0138	+0.0000 +/- 0.0000
Attack (FedAvg)	0.6928 +/- 0.0032	0.3618 +/- 0.0091	0.8825 +/- 0.0157	+0.2457 +/- 0.0023
Attack + inflation (Acc-Weighted)	0.6900 +/- 0.0047	0.3535 +/- 0.0158	0.9404 +/- 0.0101	+0.3036 +/- 0.0131
Median only	0.7098 +/- 0.0038	0.4993 +/- 0.0143	0.7009 +/- 0.0084	+0.0641 +/- 0.0064
Trust only	0.7114 +/- 0.0017	0.5311 +/- 0.0137	0.6404 +/- 0.0103	+0.0037 +/- 0.0042
Full defense (trust + median, D2)	**0.7142 +/- 0.0027**	**0.5546 +/- 0.0177**	**0.6114 +/- 0.0315**	**-0.0253 +/- 0.0178**
Full defense vs attack + inflation	0.7142 +/- 0.0027	0.5546 +/- 0.0177	0.6114 +/- 0.0315	-0.0253 +/- 0.0178

Table 2. Ablation, mean +/- std over seeds 42, 7, 123 (n = 3). BSR is the backdoor success rate on the triggered test set; backdoor lift is BSR minus the same seed's honest baseline. The highlighted row is the adopted configuration.

Insight

- The attack works and inflation makes it worse: lift +0.2457 under FedAvg, +0.3036 when fake accuracy reports buy the attackers extra weight.
- Median only helps but leaves a real backdoor effect: +0.0641 +/- 0.0064, about a quarter of the attack survives. Two attackers in ten can still drag a coordinate-wise median.
- Trust only removes nearly all of the lift (+0.0037 +/- 0.0042) because the probes identify the attackers directly. On the lift metric alone it is statistically close to the full defense, and we say so.
- The full defense is the only configuration that drives lift negative (-0.0253 +/- 0.0178) and it posts the best clean accuracy (0.7142) and spoofing recall (0.5546) at the same time. The median layer is the safety net if the trust scores are ever wrong in a round, a failure mode trust-only cannot bound.

Robustness check (last row): against attack + inflation the full defense produces numerically identical results, and the reason is structural rather than lucky. Trust comes entirely from server-side probing; the self-reported accuracy that inflation manipulates is never consulted by the full defense.

3. Trigger generalization, including a mixed trigger

Four trigger settings, each run undefended and under the full defense at the identical D2 configuration; nothing is retuned per trigger. PD is the weakest probed feature and the genuine stress test; the mixed CN0+TCD trigger is a pattern the defense was never designed around.

Trigger	Cohen's d	Attack BSR	Attack lift	Defended BSR	Defended lift	Honest FP rate
CN0	0.288	0.8825 +/- 0.0157	+0.2457 +/- 0.0023	0.6114 +/- 0.0315	-0.0253 +/- 0.0178	0.3%
TCD	0.302	0.8428 +/- 0.0075	+0.2384 +/- 0.0729	0.5419 +/- 0.0261	-0.0626 +/- 0.0512	0.7%
PD	0.152	0.9168 +/- 0.0069	+0.0909 +/- 0.0106	0.8071 +/- 0.0175	-0.0188 +/- 0.0150	0.3%
CN0+TCD	0.288 + 0.302	0.9816 +/- 0.0051	+0.1933 +/- 0.0664	0.7600 +/- 0.0683	-0.0282 +/- 0.0173	0.7%

Table 3. Trigger generalization at fixed D2, mean +/- std over seeds 42, 7, 123 (n = 3). BSR and lift are measured on each setting's own triggered test set against the same per-seed honest model. Honest FP rate is the fraction of honest client-rounds with trust below half of uniform.

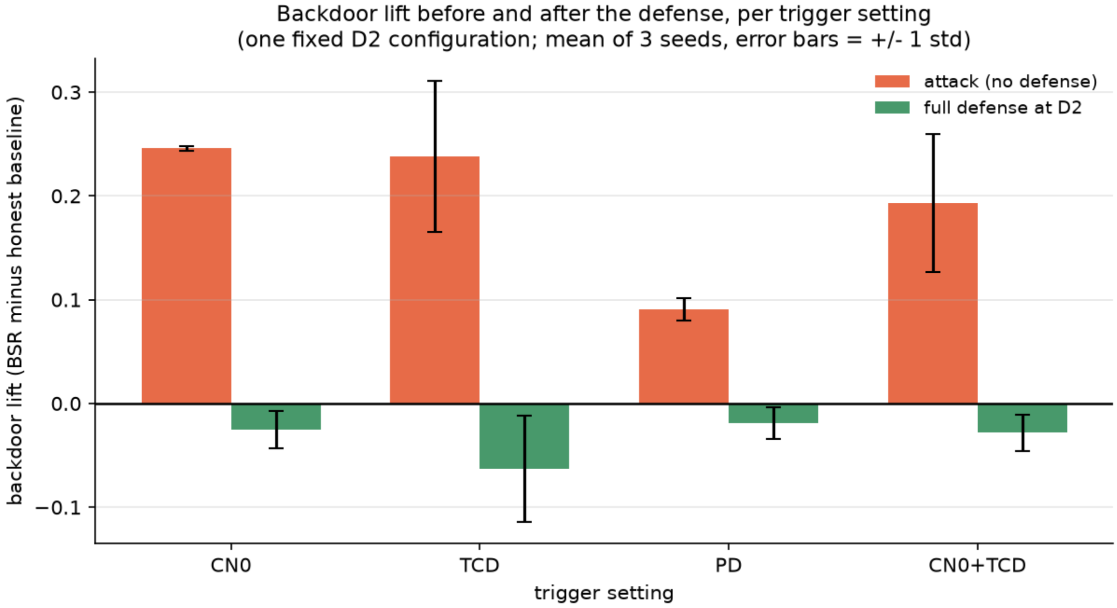

Figure 1. Backdoor lift before (orange) and after (green) the defense, per trigger setting; bars are means over three seeds with +/- 1 std error bars, per the assignment's request. The black line at zero marks "the attacker gains nothing."

Insight

- The attack carries a viable backdoor in every setting (+0.09 to +0.25 lift), so each row is a real test.
- The defense drives lift below zero in every setting: CN0 -0.0253, TCD -0.0626, PD -0.0188, mixed -0.0282. Honest false positives stay between 0.3% and 0.7% of client-rounds throughout.
- Honest observations: the seed spread on the TCD and mixed attack rows is wide (std 0.05 to 0.07), and the mixed trigger has the highest raw BSR (0.9816) but not the highest lift, because stamping two features raises the honest baseline as well.
- Together with the Week 10 five-feature sweep, this supports the claim that the defense does not require knowing the exact trigger feature, within the discriminative (probed) feature set.

4. What remains incomplete

- No adaptive attacker. Every attacker is fixed: it poisons at a set rate and scales its update without reacting to the defense. An attacker that shapes its updates to keep probe suspicion below the dead-zone tau is untested, and the dead-zone specifically invites that strategy. Strongest open threat.
- The base detector is weak. Honest spoofing recall is about 0.53 on this simplified dataset, which is why backdoor lift against each seed's own honest baseline is the primary metric.
- Three seeds. Enough to separate the attack effect from noise, but too small a sample to resolve small differences between defended variants (for example, trust-only versus full defense on lift).
- IID, small fleet. Ten clients, two attackers, IID partitions. Non-IID data would stress honest trust spread and false positives hardest; it is the natural next deeper experiment.
- Triggers outside the probe set. PQP and PIP (Cohen's d below 0.05) are excluded from probing by design and were not tested as triggers; the trigger-agnosticism claim is scoped to the discriminative feature set.